

Assignment 1

CSE 350

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sec: 1 (A)

Question-2

(d)

$$\Delta = \frac{0-0}{2} = 0$$

(a)

$$V_R = -4V$$

$$R_f = 10K\Omega, R_1 = 40K\Omega, R_2 = 20K\Omega, R_3 = 10K\Omega$$

$$V_o = |V_R| \left(\frac{S_1}{R_1} + \frac{S_2}{R_2} + \frac{S_3}{R_3} \right) \times R_f$$

For, 001,

$$V_o = |(-4)| \left(\frac{0}{40} + \frac{0}{20} + \frac{1}{10} \right) \times 10 = 4V$$

for, 010,

$$V_o = |(-4)| \left(\frac{0}{40} + \frac{1}{20} + \frac{0}{10} \right) \times 10 = 2V$$

Using this we can determine all possible input combinations for the DAC.

(b)

$$A = \frac{16-0}{8} = 2$$

$$V_0 = 0 \text{ V}$$

$$V_1 = \frac{R/2}{8R} \times 16 = 1 \text{ V}$$

$$V_2 = \frac{R+R/2}{8R} \times 16 = 3 \text{ V}$$

$$V_3 = \frac{2R+R/2}{8R} \times 16 = 5 \text{ V}$$

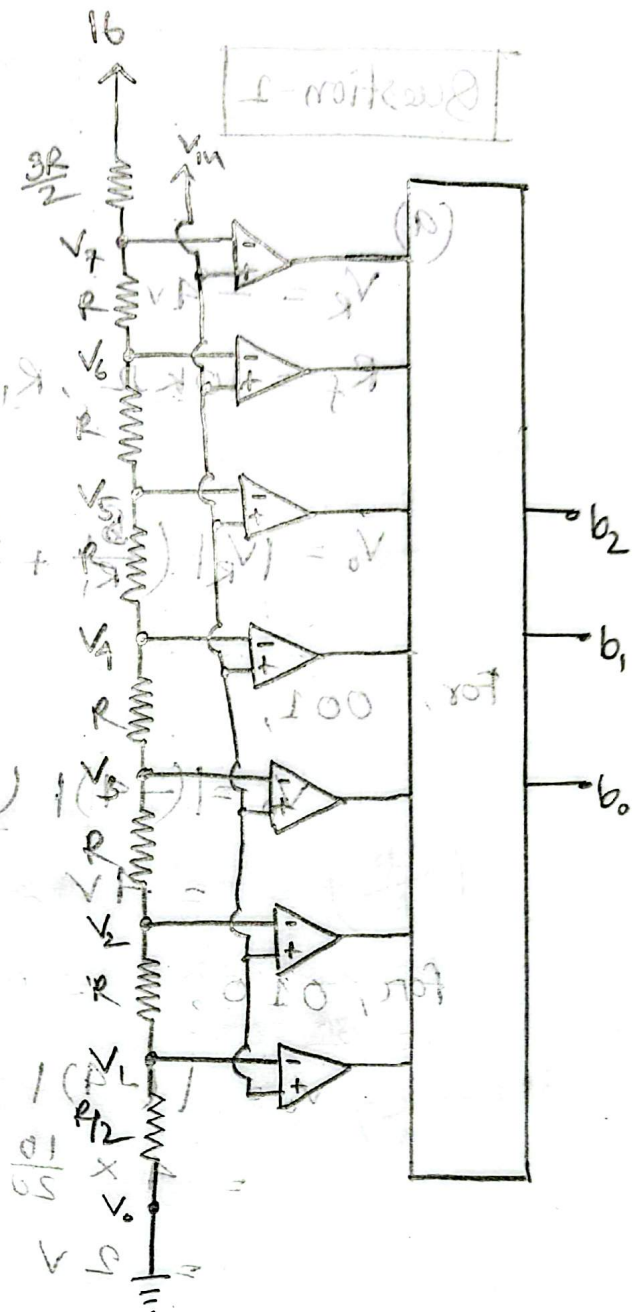
$$V_4 = \frac{3R+R/2}{8R} \times 16 = 7 \text{ V}$$

$$V_5 = \frac{4R+R/2}{8R} \times 16 = 9 \text{ V}$$

$$V_6 = \frac{5R+R/2}{8R} \times 16 = 11 \text{ V}$$

$$V_7 = \frac{6R+R/2}{8R} \times 16 = 13 \text{ V}$$

$$V_8 = 16 \text{ V}$$



output	Range (V)
000	0-1
001	1-3
010	3-5
011	5-7
100	7-9
101	9-11
110	11-13
111	13-16

For the output, 010, input voltage range is 3V-5V

(c) Range from part b is (3V-5V)

$$V_o = |V_R| \left(\frac{S_1}{R_1} + \frac{S_2}{R_2} + \frac{S_3}{R_3} \right) \times R_f$$

Using this we can determine all possible ~~com~~ input combinations for the DAC,

input	V_0 (V)
000	0
001	4
010	2
011	6
100	1
101	5
110	3
111	7

Within the range $3V \leq V_{in} < 5V$, There are two possible inputs. They are, $001 \rightarrow 4V$
 $110 \rightarrow 3V$

(range from 3 to 5V)

$$V_0 = \frac{1}{V_1} \left(\frac{R_1}{R_2} + \frac{R_2}{R_3} + \frac{R_3}{R_4} \right) \times V_1$$

Using this we can determine all possible input combinations for the DAC.

Question: 2

(a)

$$\Delta = \frac{V_{ref} - (-V_{ref})}{2^3} = \frac{2V_{ref}}{8} = \frac{V_{ref}}{4}$$

$$V_3 = -4V$$

$$V_0 = -V_{ref}$$

$$V_1 = -V_{ref} + \frac{V_{ref}}{4}$$

$$V_2 = -V_{ref} + \frac{2V_{ref}}{4}$$

$$V_3 = -V_{ref} + \frac{3V_{ref}}{4}$$

$$\therefore -V_{ref} + 3 \cdot \frac{V_{ref}}{4} = -4$$

$$\Rightarrow -V_{ref} \left(1 - \frac{3}{4}\right) = -4$$

$$\Rightarrow V_{ref} \cdot \frac{1}{4} = 4$$

$$\Rightarrow V_{ref} = 16V$$

$$\therefore -V_{ref} = -16V$$

$$V_0 = -16V$$

$$V_1 = -16 + \frac{16}{4} = -12V$$

$$V_2 = -16 + \frac{2 \cdot 16}{4} = -8V$$

$$V_3 = -4V$$

$$V_4 = -16 + \frac{4 \cdot 16}{4} = 0V$$

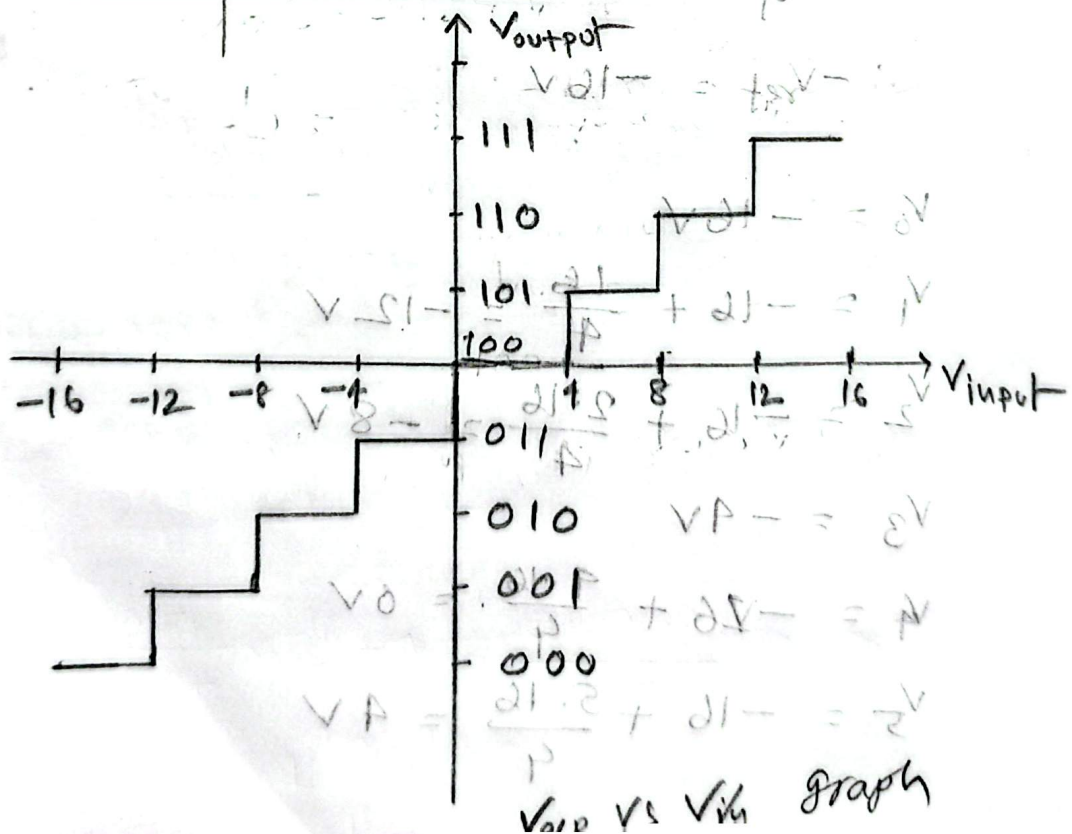
$$V_5 = -16 + \frac{5 \cdot 16}{4} = 4V$$

$$V_6 = -16 + \frac{6 \cdot 16}{4} = 8V$$

$$V_7 = -16 + \frac{7 \cdot 16}{4} = 12V$$

$$V_{ref} = 16V$$

Encoded signal	Range
000	-16 to -12
001	-12 to -8
010	-8 to -4
011	-4 to 0
100	0 to 4
101	4 to 8
110	8 to 12
111	12 to 16



(b) $V_R = 16$

Question 2

$R_f = 2 \text{ k}\Omega$

from (a), highest quantization level was ~~12V to 16V~~

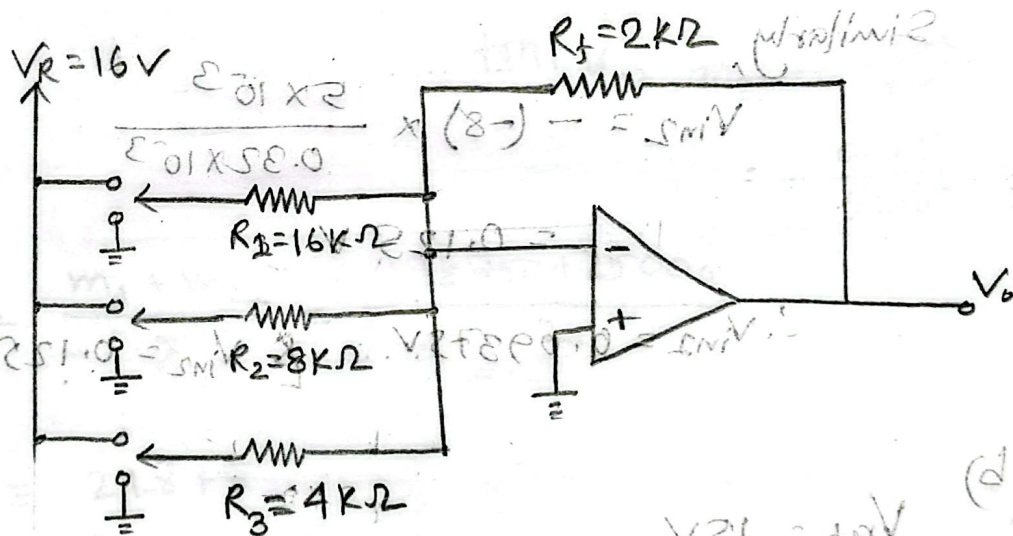
$R_3 = 2 \text{ k}\Omega$, $R_2 = 4 \text{ k}\Omega$, $R_1 = 8 \text{ k}\Omega$.

Then, maximum output (111)

$$V_o = V_R \times R_f \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} \right) = 16 \times 2 \left(\frac{7}{16} \right) = 28 \text{ V}$$

$V_o = 14$
 $V_R \cdot \frac{R_f}{R} \left(1 + \frac{1}{2} + \frac{1}{4} \right) = 14$
 $\Rightarrow R = \frac{2 \times 16 \times \frac{7}{4}}{14}$
 $\Rightarrow R = 4$

which is in the highest quantization level.



$V_{ref} = 16 \text{ V}$

Binary weighted DAC

$V_o = 14$
 $V_R \cdot \frac{R_f}{R} \left(1 + \frac{1}{2} + \frac{1}{4} \right) = 14$
 $\Rightarrow R = 4$

Question : 3

(a)

~~Q1~~

$f = 1.5625 \text{ KHz}$

$$\therefore t_1 = \frac{1}{1.5625 \times 10^3} = 0.64 \text{ ms}$$

$$V_{S1} = \left(\frac{F}{2^1} \right) \times 5 \times 21 = \left(\frac{1}{2^1} \right) \times 105 = 0.32 \text{ ms}$$

$$V_{in1} = V_1 \times \frac{R_c}{t_1} = (-6) \times \frac{5 \times 10^6}{0.32 \times 10^{-3}} = 0.09375 \text{ V}$$

Similarly,

$$V_{in2} = -(-8) \times \frac{5 \times 10^3}{0.32 \times 10^{-3}} = 0.125 \text{ V}$$

$$\therefore V_{in1} = 0.09375 \text{ V} \quad \& \quad V_{in2} = 0.125 \text{ V}$$

(b) $V_{ref} = 15 \text{ V}$,

inputs found in (a), $V_{in1} = 0.09375 \text{ V}$ & $V_{in2} = 0.125 \text{ V}$

for, V_{in2} ,

$$N_1 = \frac{V_{in2}}{V_{ref}} \times 2^6 = \frac{0.09375}{15} \times 2^8 = 1.6$$

$$= 1 \text{ (floored value)}$$

$$N_2 = \frac{V_{in2}}{V_{ref}} \times 2^8 = \frac{0.125}{15} \times 2^8 = 2.133$$

$$= 2 \text{ (floored)}$$

$$\therefore D_1 = \text{Binary of } N_1 = 00000001$$

$$D_2 = \text{Binary of } N_2 = 00000010$$

Digital outputs are D_1 & D_2 .

(c) Slope of integration for V_{in1} , $m_1 = -\frac{V_{in1}}{R_e}$

$$= -\frac{0.09375}{5 \times 10^{-6}} = -18750$$

Slope for V_{in2} , $m_2 = -\frac{V_{in2}}{R_e}$

$$= -\frac{0.125}{5 \times 10^{-6}} = -25000$$

$$-\frac{V_{in}}{R_e} = \frac{m_1 + m_2}{2} = \frac{-18750 - 25000}{2} = -21875$$

$$\Rightarrow \frac{V_{in}}{R_e} = 21875$$

$$\Rightarrow V_{in} = 0.109 \text{ V}$$

For, V_{in}

$$N = \frac{V_{in}}{V_{ref}} \times 2^8 = \frac{0.109}{15} \times 2^8 = 1.87$$

$$= 2 \text{ (floored value)}$$

$$D = \text{Binary of } N = 00000010$$

$$(10000000)_{10} = 2^7$$

$$(d) \text{ output} = 11001010$$

$$D = 202$$

Input voltage,

$$V_{in} = \frac{D}{2^n} \times V_{ref} = \frac{202}{256} \times 15 = 11.836V$$

Integrator voltage,

$$V_i = V_{in} \times \frac{t_1}{R_1 C} = 11.836 \times \frac{0.32 \times 10^{-3}}{5 \times 10^{-6}} = 757.5V$$

$$V_i = 757.5V$$

$$(e) \text{ count} = 153$$

$$\therefore V_{in} = \frac{153}{256} \times 15 = 8.965V$$

$$V_i = 8.965 \times \frac{0.32 \times 10^{-3}}{5 \times 10^{-6}} = 573.75V$$

Corresponding analog input range,

$$\frac{D}{2^n} \times V_{ref} \text{ to } \frac{D+1}{2^n} \times V_{ref}$$

$$\frac{153}{256} \times 15 \text{ to } \frac{154}{256} \times 15$$

$$\text{or } 8.965V \text{ to } 9.023V$$

Question - 4)

(a)

$$V_{ref} = -10V$$

$$R = 5k\Omega$$

$$I_1 = \frac{V_{ref}}{2R} = \frac{-10}{2 \times 5} = -1mA$$

$$I_2 = \frac{I_1}{2} = -\frac{1}{2} = -0.5mA$$

the output is inverted, so the output is positive

$$I_3 = \frac{I_2}{2} = -0.25mA$$

(b)

~~$$V_{max} = -V_{ref} \times \frac{R_f}{2R} \left(1 + \frac{1}{2} + \frac{1}{4}\right) = -(-10) \times \frac{10}{5} \left(1 + \frac{1}{2} + \frac{1}{4}\right)$$~~

$$\text{Resolution} = \frac{|V_{ref}|}{2^n} \times \frac{R_f}{R}$$

$$= \frac{|-10|}{8} \times \frac{10}{5}$$

$$= 2.5$$

(c)

$$V_{max} = -R_f (I_1 + I_2 + I_3) = -10 \times (-1 - 0.5 - 0.25) = 17.5V$$

$$V_{min} = 0V$$

(A - not given)

(d)

1 MSB (100)

$$V = -R_f(I_1) = -10 \cdot \left(\frac{1}{2}\right) = -5V$$

$$= 10V$$

$$\therefore \frac{1}{2} \text{ MSB} = \frac{1}{2} \times 10 = 5V$$

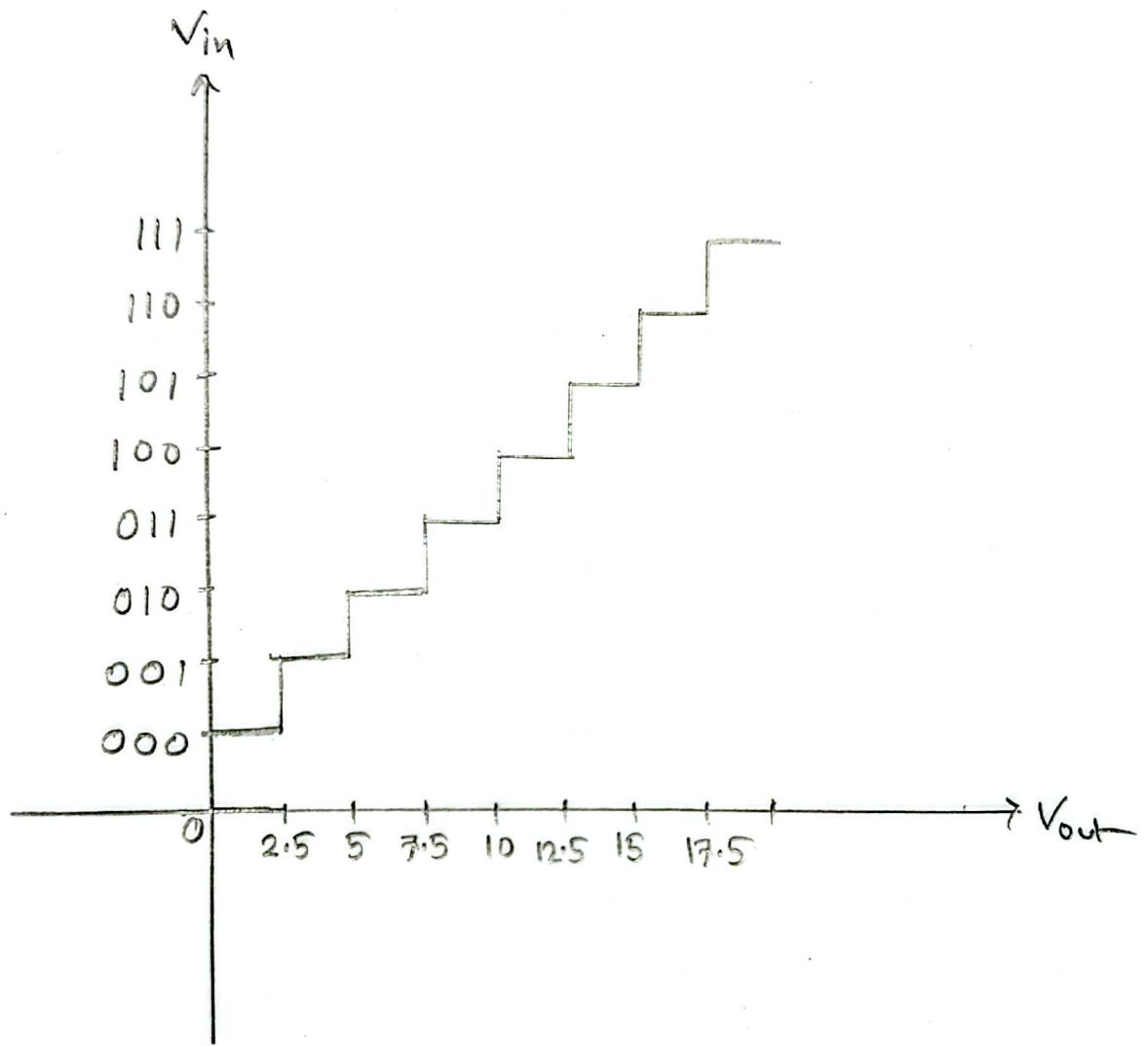
$$A_{m2.0} = \frac{1}{5} = \frac{1}{5} = 0.2$$

(e) $V_o = D \times 2.5V$, (where, D = Decimal equivalent of the input code.)

Input	Output
000	0
001	2.5
010	5
011	7.5
100	10
101	12.5
110	15
111	17.5

$$V_{ref} = 10V$$

$$V_o = 10V$$



V_{in} vs V_{out} graph